



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

USB to EtherCAT Adapter Board V.1.0

The domain of the Project:
PCB DESIGNING

Mentors NAME : MR.PARAMESH KUMAR

BY
MR. SUDESH PANDIT

Period of the project
1 Months

March 2026 to September 2026



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Declaration

The project titled “**USB to EtherCAT Adapter Board V.1.0**” has been mentored by **Mr. Paramesh Kumar Sir**, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Sudesh Pandit

----- -- Signature

Mentor's Name -Mr. Paramesh Kumar Sir
Designation—Hardware Design Engineer

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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List of PCB Projects Completed

S.no	Project Name	Drive Link
1	Project 1 Circuit	https://drive.google.com/file/d/1epLdO8WDVbeXeT19_Wi16udz81mMlt-P/view?usp=sharing
2	Project 2 - Usb to serial	https://drive.google.com/file/d/1i2XWfAvxnklXVdDYmle6WPvFxUR9vWKw/view?usp=sharing
3	Tsh82 Configurable OP-Amp Board	https://drive.google.com/file/d/1E38N-GScFRocY2bOcbLQZKqGncWNo7Cy/view?usp=sharing
4	Tsh82 Configurable OP-Amp Board_2nd	https://drive.google.com/file/d/1sDuhp4cLL0noDjEnpnJrsfTknCsBbXIY/view?usp=sharing
5	Qwiic_Power_Switch	https://drive.google.com/file/d/1XztfB4fw7R8LX9Kovd2zO7RbngG7HY7W/view?usp=sharing
6	Qwiic_Power_Switch - 2nd	https://drive.google.com/file/d/1swSz5B1-ilnt20oaJMDKc0X0MihFqqRX/view?usp=sharing
7	8051 Breakout board	https://drive.google.com/file/d/1nDGoI3qU09t9esy2-FVGr2dhfmRGa3Lm/view?usp=sharing
8	Arduino-Pro-Mini-SMD	https://drive.google.com/file/d/1IHXNaueF0L5I3LIYoW2jHVUwuduZDFr/view?usp=sharing
9	Arduino-Pro-Mini-THT	https://drive.google.com/file/d/1rmi6C_UNqyQRsobWJNSHu1F2Q-eT9DB/view?usp=sharing



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10	Arduino_MCU	https://drive.google.com/file/d/1RZyvrOXFr43jHNar_mCFisNDPLKOBTez/view?usp=sharing
11	Arduino_MCU_Datalog	https://drive.google.com/file/d/1PX1Dq6tHUcxDC9c_TriYwdU33XxcEk0J/view?usp=sharing
12	STM32F1 _Dev_Board	https://drive.google.com/file/d/1iwejlBlvHLYP5TMjYkbf1mXFZMGVwR3/view?usp=sharing
13	STM32F1_Dev_Board- 2	https://drive.google.com/file/d/1T_0eOOVG7BfFlnb_VGHcRWAqw7xLhs6r/view?usp=sharing
14	USB to Ethercat Adapter Board V.1.0	https://drive.google.com/file/d/1IbJHFMyg_S0Z5MWAGBCIC6oQwrj6rXUx/view?usp=sharing



1. Executive Summary

Objectives: This project set out to design, lay out and fabricate a two-layer PCB — the USB to EtherCAT Adapter Board V1.0 — in KiCad, bridging a standard USB port on a PC to an EtherCAT industrial network. The board combines an STM32F411CEUx (ARM Cortex-M4) microcontroller with a Microchip LAN9252 two-port EtherCAT slave controller, an AMS1117-3.3 regulator, a 24LC512 I2C configuration EEPROM, and two RJ45 ports (IN/OUT) for EtherCAT daisy-chaining, all on a compact two-layer FR-4 substrate.

Methods: The complete PCB design workflow was followed solo, end-to-end, in KiCad — schematic capture, footprint/component placement, two-layer routing, application of industry-standard design rules (trace width, clearance, via/annular ring, pour clearance, silkscreen clearance, edge clearance), a full Design Rule Check (DRC), and generation of fabrication-ready Gerber/Excellon files, followed by assembly and functional bring-up testing.

Key Findings: The finished design passed DRC with 0 violations and 0 unconnected items. Tracing every +5V net on the schematic showed it reaches only two points (the USB input and the regulator input), while the +3.3V rail correctly supplies every active IC — confirming clean power-domain separation. On the assembled board, USB enumeration, SPI communication between the STM32 and the LAN9252, and link-up on both RJ45 ports were all verified working.

Recommendations: For a production version, migrating to a 4-layer stack-up with dedicated power/ground planes and adding stronger ESD/EMI protection on the USB and RJ45-facing lines is recommended to improve signal integrity and industrial-environment robustness (see Section 7.2 for the full future-scope list).

This report is intended to give stakeholders and reviewers a complete, self-contained picture of the board's function and design quality — the full portfolio of 14 PCB projects completed during this training is listed for reference right after the Table of Contents, with this adapter board described in full detail from Section 1 onward.



2. Introduction

2.1 Background and Context

Printed Circuit Boards (PCBs) are the backbone of nearly every modern electronic product — they mechanically hold components in place and provide the copper pathways that carry power and signals between them. As part of the SURE Trust hardware and embedded systems training, the trainee was required to design, lay out and fabricate a real, functioning 2-layer PCB in KiCad rather than working only with breadboards or perfboards, so that the complete journey from schematic to a manufacturable board is experienced first-hand.

2.2 Problem Statement / Goals

EtherCAT is a widely used industrial real-time Ethernet fieldbus, but most PC/laptop hosts only expose a USB port and cannot join an EtherCAT network directly. A dedicated adapter is needed that a PC can talk to over standard USB, while the board itself carries the silicon needed to actually speak EtherCAT on the wire — an EtherCAT slave/communication controller with its own dual Ethernet PHYs and RJ45 daisy-chain ports. The goal was to design a compact, USB-powered 2-layer PCB combining an STM32 microcontroller with a Microchip LAN9252 EtherCAT controller that performs this bridging role, while giving the trainee hands-on exposure to two-layer PCB layout and manufacturing design rules in KiCad.

2.3 Scope and Limitations

- **Scope:** schematic design in KiCad, 2-layer PCB layout, layer-wise documentation, DRC verification, and functional testing of one production-representative board.
- **Limitation:** the board is a single prototype/pilot run; enclosure design and mass-production tooling (stencils, panelisation) were out of scope for this training cycle.

2.4 Innovation Component

Rather than using two separate ready-made modules, the STM32F411 microcontroller, the LAN9252 EtherCAT controller with its dual integrated Ethernet PHYs, the I2C configuration EEPROM, USB interface and power regulation were combined onto one custom 2-layer PCB. The PHY power-supply domains (VDD12TX, VDD33TXRX1/2) are each individually filtered with ferrite beads and local decoupling — a practical, industry-style design decision, explained later in this report, that keeps the high-speed Ethernet PHY signals clean on a low-cost 2-layer stack-up.



3. Project Objective

3.1 Clearly Defined Objectives and Goals

- Design a complete KiCad schematic combining an STM32F411CEUx microcontroller with a Microchip LAN9252 EtherCAT slave controller, dual RJ45 EtherCAT ports, an I2C configuration EEPROM, USB host interface, and 5V-to-3.3V regulation.
- Lay out the design on a 2-layer PCB, using the top layer primarily for signal/component routing and the bottom layer primarily as a ground plane, with careful attention to the Ethernet PHY power-supply filtering.
- Apply and document standard PCB design rules (trace width, clearance, via/annular ring sizing, pour clearance, silkscreen clearance, edge clearance) suitable for a low-cost fabrication house, and verify them with a full Design Rule Check (DRC).
- Clearly document every physical layer and view of the PCB (top/bottom copper, front/back assembly, 3D preview) with labelled images.
- Explain, block by block, how the actual KiCad schematic works, and highlight every +5V and +3.3V net directly on the schematic drawing.
- Fabricate, assemble and functionally test the board, confirming stable power rails, correct USB enumeration, correct SPI communication between the STM32 and the LAN9252, and Ethernet link-up on both RJ45 ports.

3.2 Expected Outcomes and Deliverables

- A DRC-clean, fabrication-ready 2-layer PCB design (schematic + layout) for the USB to EtherCAT Adapter Board V1.0, exported as manufacturing-ready Gerber/Excellon files.
- A fully documented design record: layer stack-up, top/bottom copper routing, front/back assembly views, a 3D pre-fabrication preview, and every design rule applied — usable as a template for future board designs.
- A working, tested prototype board confirmed over USB enumeration, SPI communication with the LAN9252, and Ethernet link-up on both RJ45 ports.
- This report itself, intended as a learning resource that walks a reader through the complete PCB design process from schematic to fabrication in the order the work was actually done.



4. Methodology & Results

4.1 Methods / Technology Used

- Schematic capture and PCB layout carried out entirely in KiCad 10.0.0 (Eeschema for schematic, Pcbnew for layout), following the workflow: schematic → footprint assignment → PCB layout → DRC → Gerber export.
- 2-layer FR-4 substrate, 1.6 mm thickness, 1 oz copper per layer — the most common and cost-effective stack-up for prototype and small-batch boards.
- STM32F411CEUx (ARM Cortex-M4, UFQFPN48) as the host-side microcontroller, communicating with the EtherCAT controller over SPI.
- Microchip LAN9252 — a two-port EtherCAT slave controller (ESC) with two integrated 100Base-TX Ethernet PHYs — as the dedicated EtherCAT communication IC.
- AMS1117-3.3 linear regulator to step the USB 5V down to a clean 3.3V rail for the microcontroller, the EtherCAT controller and the EEPROM.

4.2 Tools / Software Used

- KiCad EDA suite 10.0.0 — Schematic Editor, PCB Editor and the built-in 3D Viewer (as seen in the project screenshots in this report)
- Design Rule Check (DRC) and Electrical Rule Check (ERC) built into KiCad
- Gerber/Excellon viewer to verify fabrication files before sending them to the manufacturer
- ST-Link / SWD debug probe connected via header J5 for firmware flashing and debugging
- Multimeter and continuity tester for post-assembly bring-up and rail verification

4.3 Data Collection Approach

Not applicable in the traditional sense — the "data" collected for this project consists of design verification data (DRC/ERC reports and continuity test results) and functional bring-up data (USB enumeration status, SPI communication with the LAN9252, and RJ45 link-status LEDs), which is summarised in Section 4.6.

4.4 Project Architecture

The board follows a bridge architecture: a USB connector (J3) supplies +5V and carries the USB D+/D- data lines into the STM32's native USB pins (PA11/PA12), giving a PC direct USB access to the board. The STM32 (U1) talks to the LAN9252 EtherCAT controller (U4) over an



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SPI bus (SCS, SCK and data lines), and the LAN9252 in turn drives its two integrated Ethernet PHYs out to the RJ45 IN (J2) and RJ45 OUT (J1) connectors, which provide the physical EtherCAT network ports used to daisy-chain the adapter with other EtherCAT devices. A 24LC512 I2C EEPROM (U2) attached to the LAN9252's I2C pins stores the mandatory EtherCAT Slave Information Interface (SII) configuration data that the LAN9252 reads on power-up.

The complete schematic below is a direct screenshot from the KiCad Schematic Editor for this project. It is organised into functional blocks arranged left to right, top to bottom: the STM Microcontroller (top-left), the LAN9252 EtherCAT controller with its PHY power filtering (top-right), the RJ45 Connector IN (mid-left), the RJ45 Connector OUT (bottom-left), the USB block and the 5V-to-3V3 regulator (middle), the I2C EEPROM (bottom-middle), and the mounting holes plus title block (bottom-right).

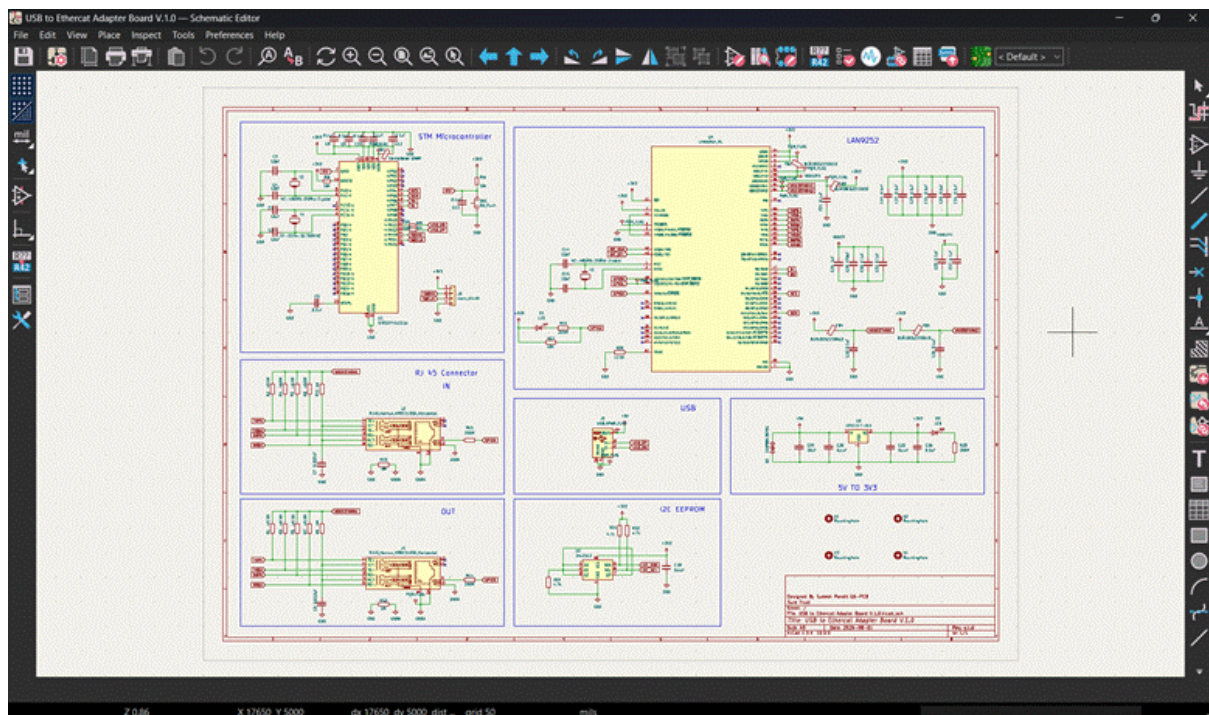


Figure 1: Complete KiCad schematic — USB to EtherCAT Adapter Board V1.0 (STM32F411 + LAN9252).

How the Circuit Works — Block-by-Block

Reading the schematic block by block, the design works as follows:

STM Microcontroller (U1 – STM32F411CEUx): The host-side processor. Its VBAT/VDD/VDDA pins are decoupled with a bank of 0.1 μ F capacitors (C8–C12) and isolated from the analog supply by ferrite bead FB1. A 25 MHz crystal (Y2, with C3/C4) drives its main oscillator, while a 32.768 kHz watch crystal (Y1, with C1/C2) drives the RTC. NRST has a manual reset button (SW1) with pull-up R16, and the chip's native USB pins (PA11/PA12) connect straight to the USB connector. Four SPI-related GPIOs (SCS, SCK, and two data lines) connect to the LAN9252, and SWDIO/SWCLK are broken out to the SWD header J5 for programming.



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LAN9252 EtherCAT Controller (U4): A Microchip LAN9252 — a two-port EtherCAT Slave Controller (ESC) with two integrated 100Base-TX Ethernet PHYs. It is configured over SPI by the STM32 (SCS chip-select, SCK clock, and data lines) and reads its mandatory device configuration (Vendor ID, Product Code, PDI settings) from the external I2C EEPROM on power-up. Its internal power domains — VDDIO/VDDCR (digital), VDD12TX (PHY core, 1.2 V), and VDD33TXRX1/VDD33TXRX2 (PHY transceivers for ports A and B) — are each fed from the +3.3V rail through individual ferrite beads (FB2–FB5) with local bulk and bypass capacitors, isolating the sensitive analog PHY circuitry from digital switching noise elsewhere on the board.

RJ45 Connector IN / OUT (J2, J1): Two RJ45 jacks with integrated magnetics (HR911105A) connect the LAN9252's two internal PHYs (Port A → RJ45 IN, Port B → RJ45 OUT) to the outside world. Each port has a 49.9 Ω termination-resistor network (R1–R8) biasing the differential TD/RD pairs and an AC-coupling capacitor (C7) to a floating centre-tap — the standard 100Base-TX magnetics reference circuit. Having two independent ports lets multiple EtherCAT devices be daisy-chained: EtherCAT traffic comes IN on one port and continues OUT to the next device in the network. Each jack's integrated LEDs are driven from LAN9252 GPIOs (via R14/R15) to show per-port link/activity status.

USB Interface (J3): A USB connector supplies +5V on VBUS and carries USB_DP/USB_DN straight into the STM32's native USB peripheral. This is the link a host PC uses to talk to the board — for firmware/configuration traffic and, depending on firmware, to relay EtherCAT process data — without needing any external USB-to-serial chip.

5V to 3V3 Regulation (U5 – AMS1117-3.3): USB VBUS first passes through TVS diode D2 (ESD9B5.0STDG) for ESD/surge protection, then into the AMS1117-3.3 regulator, bulk-filtered by C19 (10 μ F) on the input and C22/C26 (0.1 μ F) on the output. The resulting +3.3V rail powers the STM32, the LAN9252's digital domains, and the EEPROM. LED D3 (with R25) lights whenever +3.3V is present, acting as a simple power-good indicator.

I2C EEPROM (U2 – 24LC512): A 512 Kbit I2C EEPROM with its address pins (A0–A2) tied to GND (address 0x00) and pull-ups R21/R22 on SDA/SCL. This chip stores the LAN9252's EtherCAT Slave Information Interface (SII) data — the Vendor ID, Product/Revision codes and PDI configuration that the LAN9252 automatically loads over I2C every time the board powers up, which is what lets an EtherCAT master correctly identify and configure this device on the network.



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4.5 The PCB Design Journey — From Schematic to Fabrication

With the schematic finished and electrically verified, the design moves from Eeschema into Pcbnew for the physical PCB layout. The steps below are presented in the same order the board was actually built — starting from the basic building blocks of a PCB, through component placement, copper routing, the design rules applied along the way, via usage, net verification, and the final Design Rule Check — so that a reader new to PCB design can follow the whole journey end to end.

Step 1 — Understanding the Parts of a PCB

Before laying anything out, it helps to know the physical building blocks that make up any PCB, all of which are present on this 2-layer design:

Substrate (FR-4 Core): The rigid fibreglass-epoxy base material that gives the board its mechanical strength and acts as the insulator between the two copper layers.

Copper Layer: A thin layer of copper laminated onto the substrate and etched into the traces, pads and planes that carry electrical signals and power.

Solder Mask: The green (or coloured) lacquer coating that covers copper areas which should not be soldered, preventing bridges and oxidation.

Silkscreen: The white ink layer printed on top of the solder mask carrying component reference designators, polarity marks, logos and labels.

Pads: Exposed copper areas shaped to match a component's leads/pins where solder physically and electrically connects the part to the board.

Traces (Tracks): Copper paths that route electrical signals and power between pads, sized according to the current they must carry.

Vias: Plated through-holes that electrically connect copper on one layer to the other layer, allowing traces to change layers.

Ground/Power Plane (Copper Pour): A large flooded copper area (usually on the bottom layer) tied to GND that lowers impedance, improves EMI performance and shields signals.

Mounting Holes: Plated or unplated holes at the board corners used to mechanically secure the PCB inside an enclosure.

Board Outline (Edge Cut): The mechanical boundary defining the physical shape and size of the finished, routed PCB.

Step 2 — Planning the Layer Stack-Up

The very first layout decision is how many copper layers the board will have and how they are stacked. This design uses a standard, low-cost 2-layer (double-sided) stack-up.



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2-Layer PCB Stack-Up (Cross-Section View)

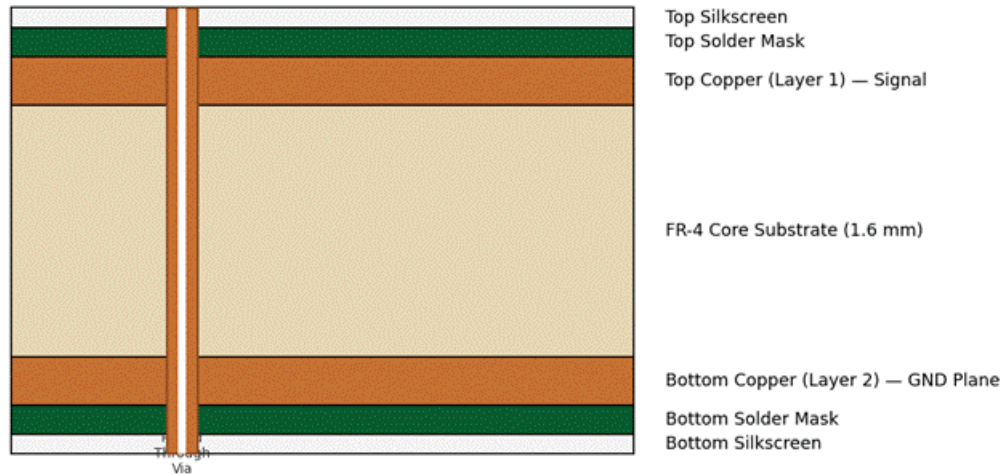


Figure 2: Cross-section of the 2-layer stack-up — copper on the top and bottom of a single FR-4 core, each protected by solder mask and silkscreen.

The stack-up shows the two copper layers separated by a 1.6 mm FR-4 core, with a plated through-hole via connecting both copper layers, and solder mask/silkscreen applied on both outer faces. This decision — 2 layers, not 4 — is what makes the ground-pour strategy in later steps so important.

Step 3 — Component Placement: Front Assembly View

With the stack-up fixed, every component's footprint was placed on the board outline. The front (top) assembly view below is a direct KiCad screenshot showing every component outline and reference designator on the component side.

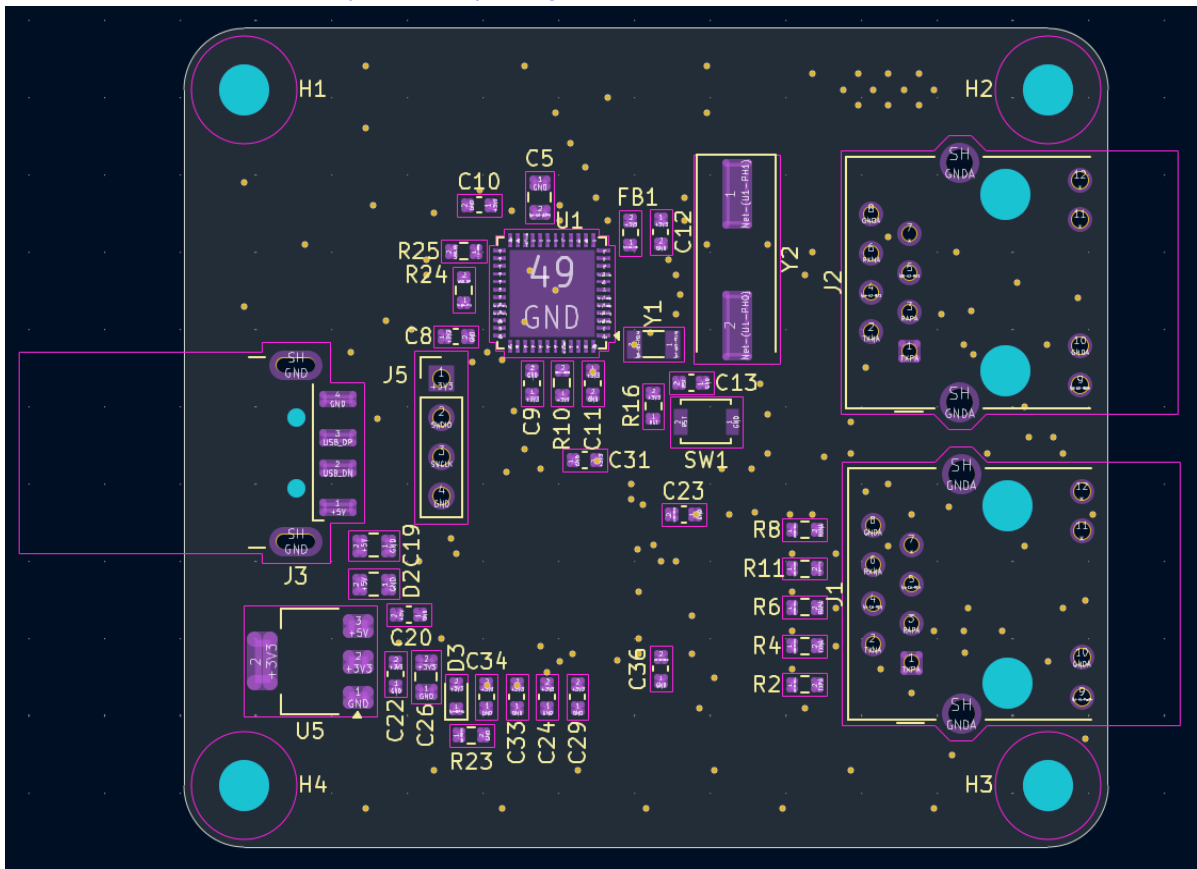


Figure 3: Front Assembly View (KiCad Pcbnew) — U1 (STM32), Y1/Y2 crystals, J3 (USB), J2/J1 (RJ45), SW1, and the passive components on the top side.

This is the actual KiCad Pcbnew screenshot for this board (Front Assembly View preset). Placement decisions here — putting the STM32 centrally, the USB connector at the board edge, and both RJ45 jacks along the opposite edge — directly shaped how short and clean the routing could be in the next steps.

Step 4 — Component Placement: Back Assembly View

This design places no components on the bottom side — it is a single-sided assembly — but the back view is still a real, useful layer: it shows every through-hole component's footprint outline as seen from below, plus the LAN9252 and its surrounding passives silhouetted through the board.

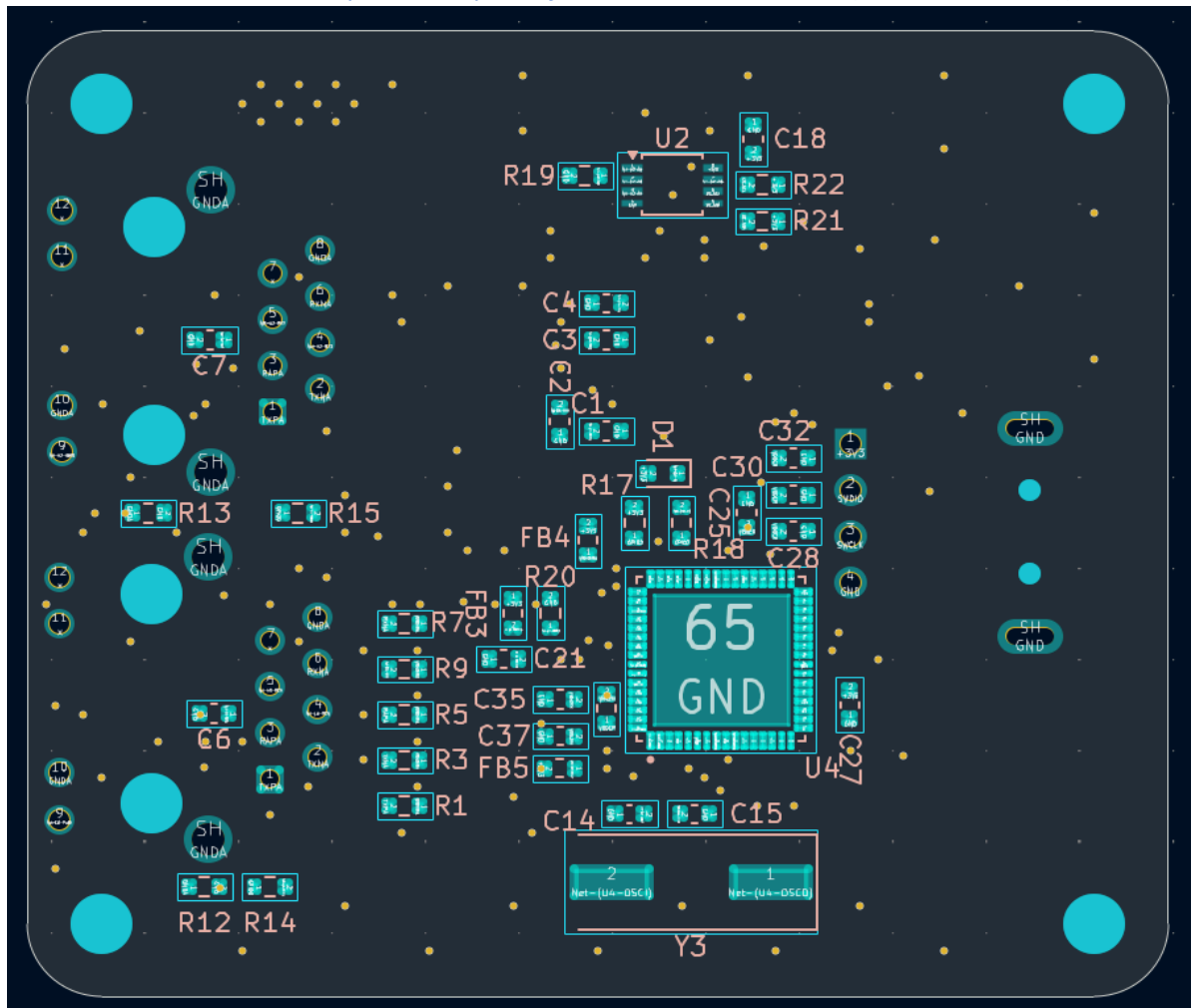


Figure 4: Back Assembly View (KiCad Pcbnew) — component outlines as seen from the bottom side, including the LAN9252 (U4) footprint.

Reviewing both the front and back assembly views together before routing helped catch any footprint overlap or spacing issue early, while components could still be moved freely.

Step 5 — Routing the Top Copper Layer (Front)

Routing began on the top copper layer (F.Cu, shown in red), connecting the STM32's SPI bus to the LAN9252, the USB data lines, and the Ethernet PHY differential pairs out to both RJ45 connectors.

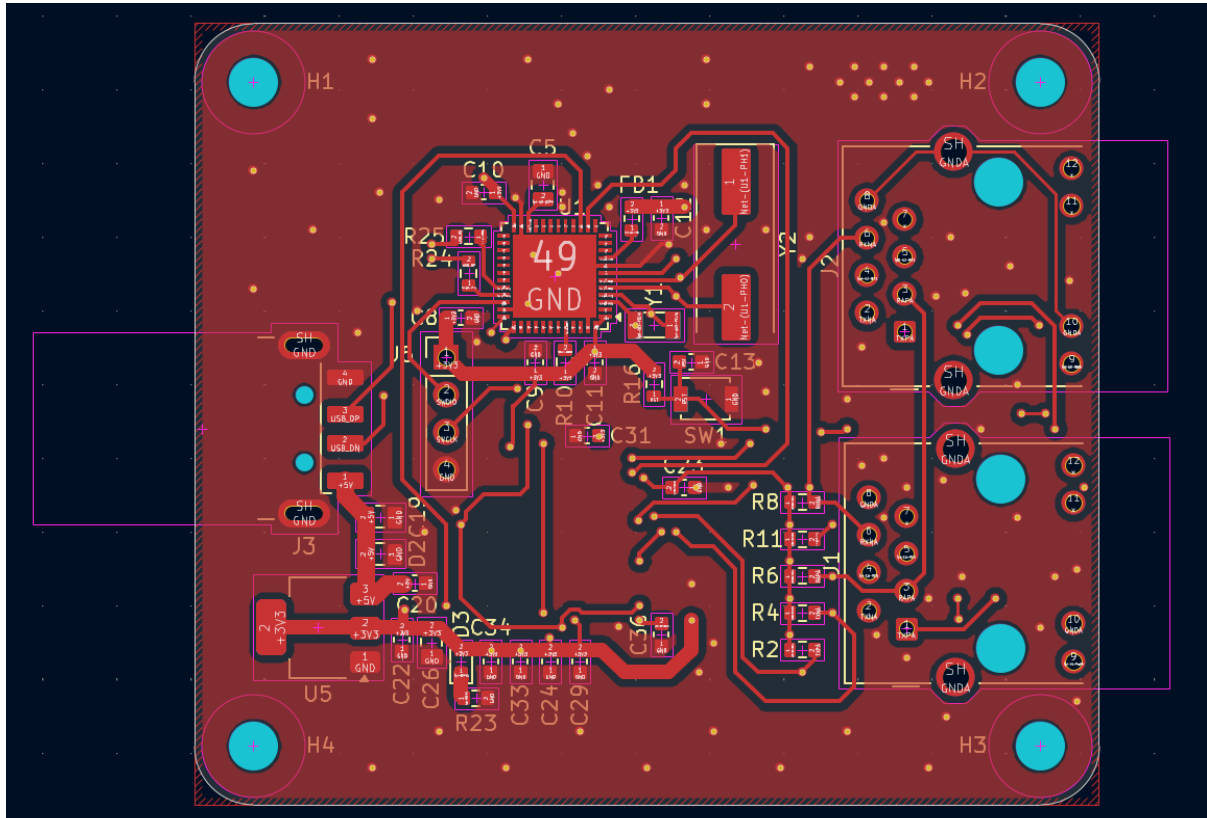


Figure 5: Top Copper Layer (F.Cu) — actual routed layout from KiCad Pcbnew, Front Layers view.

This is the real routed top layer for this board — visible here are the SPI traces running from the STM32 ("49 / GND" QFN, left) to the LAN9252 ("49" pad marked, right), the USB traces to J3, and the differential pairs fanning out to the RJ45 headers J1/J2 on the right edge.

Step 6 — Routing the Bottom Copper Layer (Back / Ground Plane)

The bottom copper layer (B.Cu, shown in blue) was used mainly as a solid ground pour beneath the LAN9252 and its Ethernet PHYs — the most EMI-sensitive part of the board — with a handful of jumper traces routed here only where the top layer became too congested.

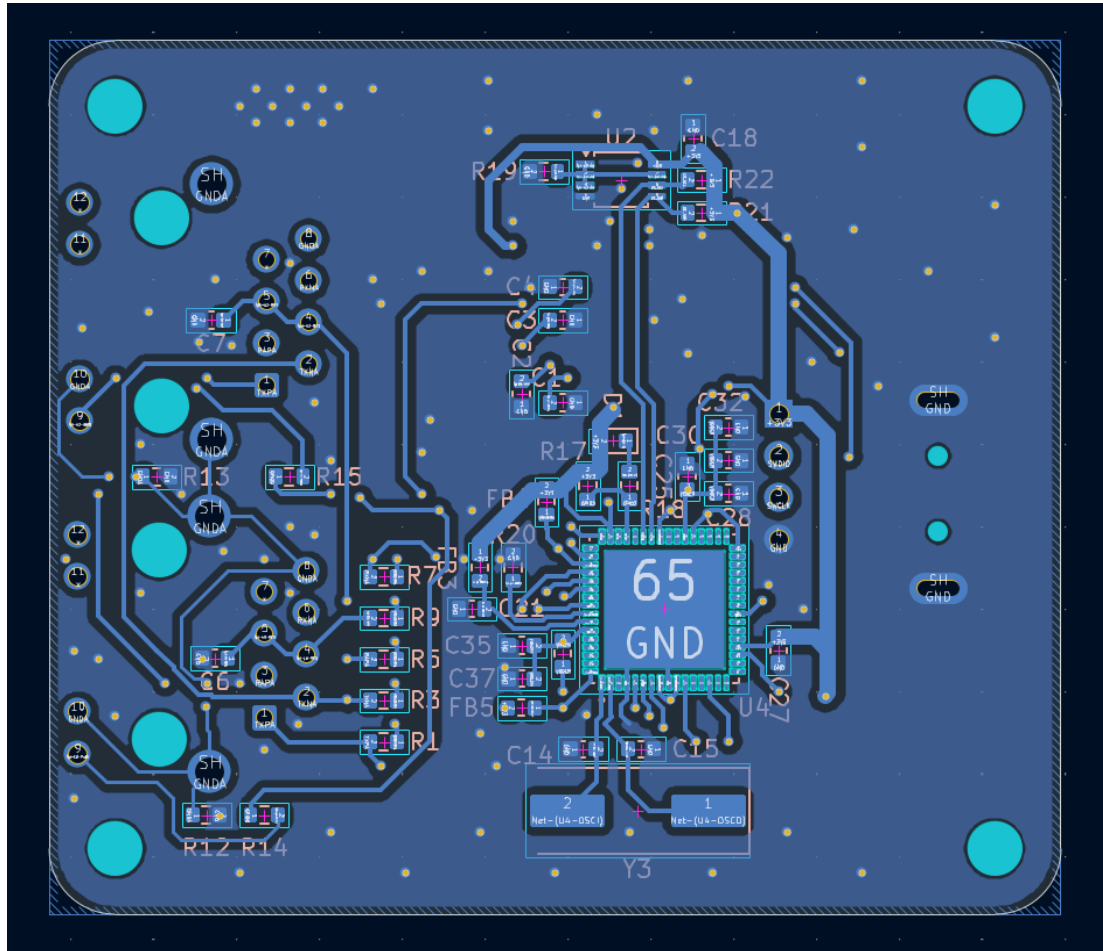


Figure 6: Bottom Copper Layer (B.Cu) — actual routed layout from KiCad Pcbnew, Back Layers view, showing the ground pour and LAN9252 ("65 / GND") footprint.

Keeping this layer mostly as unbroken ground copper — rather than routing lots of signal traces here — was a deliberate choice to give the Ethernet PHY signals on the top layer a clean, continuous return path directly beneath them.

Step 7 — Connecting Layers with Vias

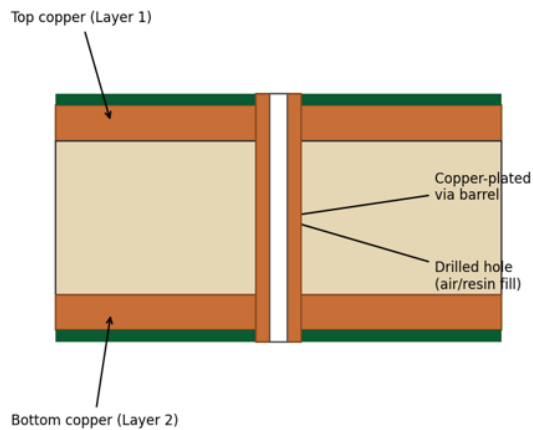
Wherever a signal needed to move from the top copper layer to the bottom layer (or vice versa) — for example, jumping a trace under a component or dropping down to the ground pour — a via was used. A via is a small plated hole that carries the copper connection through the board.



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Via Concept — Electrically Connecting Layer 1 and Layer 2

Cross-Section: Through-Hole Via



Top-Down View: Via Pad, Drill & Annular Ring

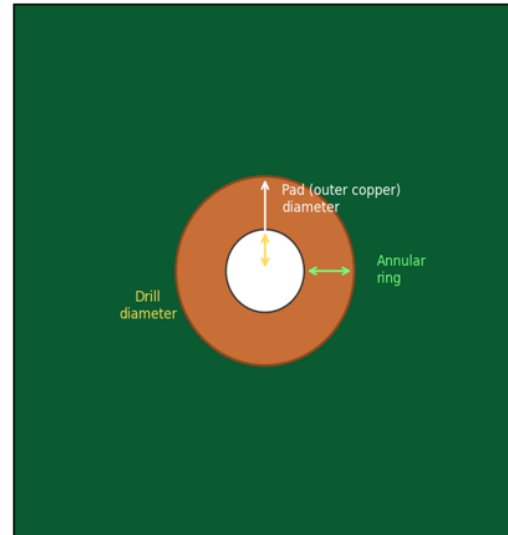


Figure 7: Via concept — cross-section of a plated through-hole via, and a top-down view showing drill diameter, pad diameter and annular ring.

The left-hand cross-section shows the copper-plated barrel running through the FR-4 core between the top and bottom copper layers; the right-hand top-down view shows the via's key dimensions — drill diameter, pad diameter, and the annular ring (copper ring left around the hole), which is exactly what Design Rule 2 below checks. On multilayer (4-layer+) boards, blind and buried vias are also possible, but this 2-layer design uses only simple through-hole vias, visible as the small copper circles throughout Figures 5 and 6.

Step 8 — Applying PCB Design Rules

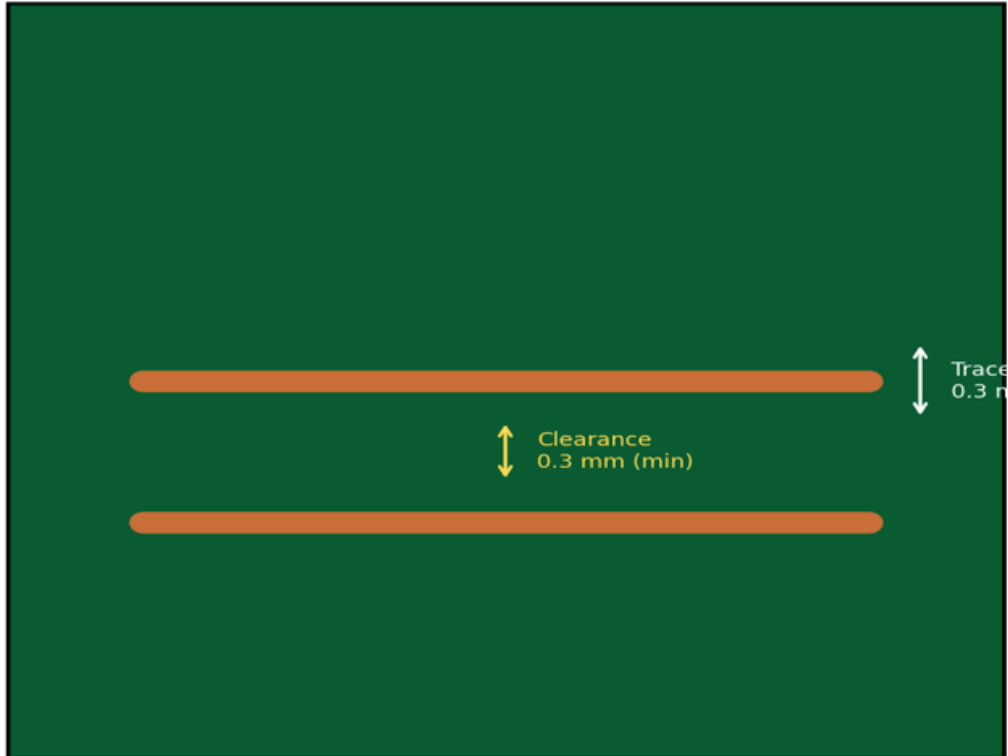
While routing, a set of manufacturing design rules was followed continuously — not just checked at the end — so that every trace, pad and via was placed within safe limits for a low-cost fabrication house from the very first click. Each rule is illustrated below with an annotated diagram.



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Rule 1 — Trace Width & Trace-to-Trace Clearance

Design Rule 1: Trace Width & Trace-to-Trace Clearance



Rule 1 — Trace Width & Trace-to-Trace Clearance — annotated design-rule illustration.

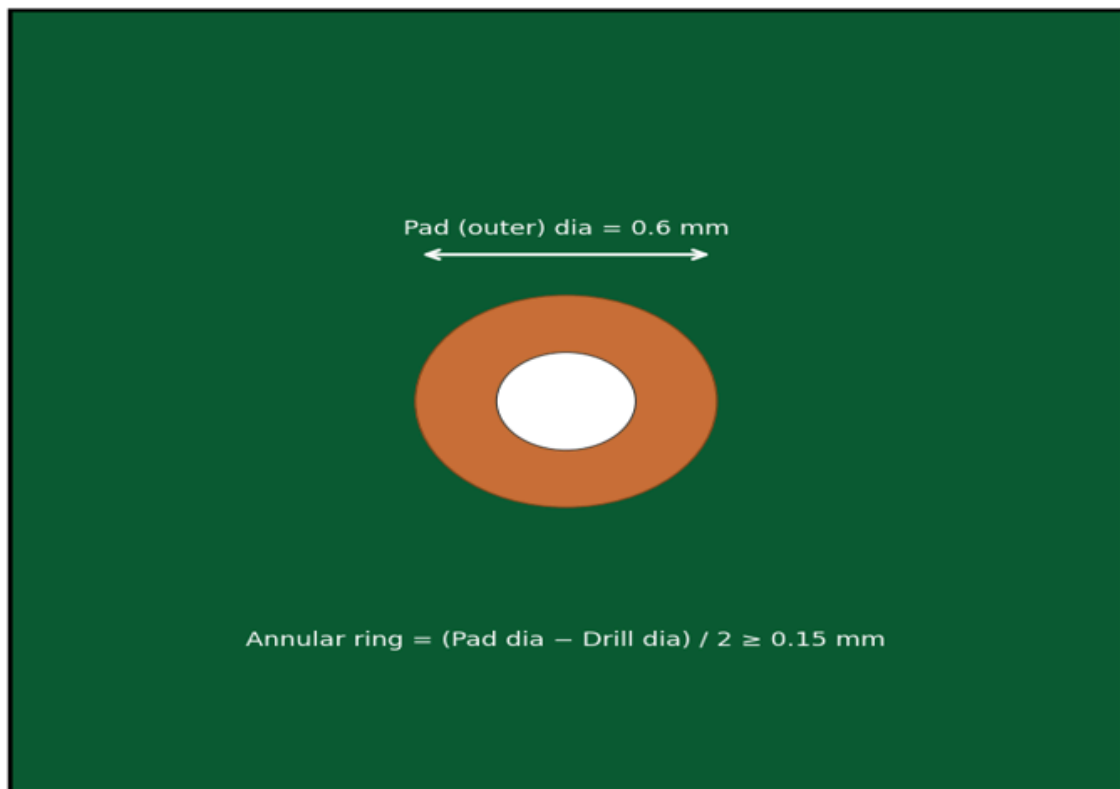
A minimum trace width of 0.3 mm was used for signal lines and a minimum clearance of 0.3 mm was maintained between adjacent copper features to avoid short circuits during etching and soldering.



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Rule 2 — Via Drill Size & Annular Ring

Design Rule 2: Via Drill Size & Annular Ring



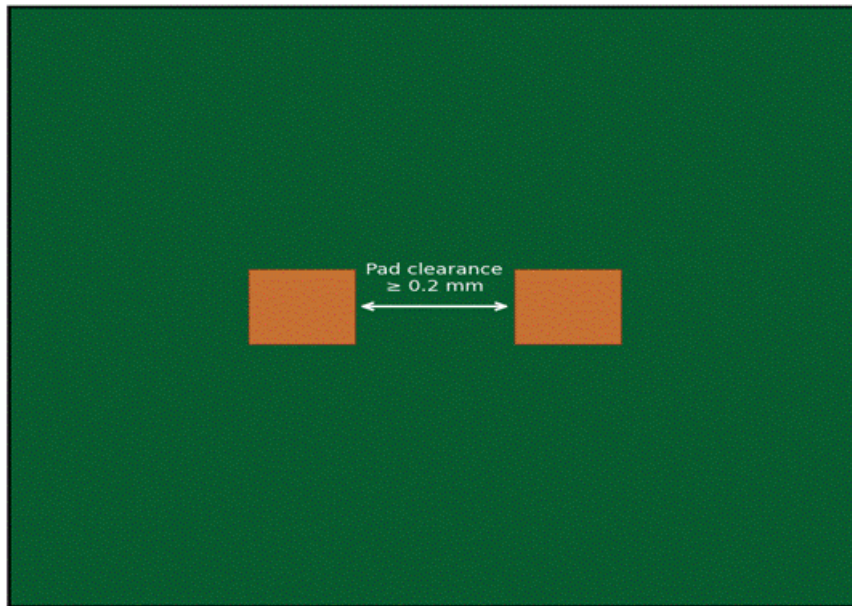
Rule 2 — Via Drill Size & Annular Ring — annotated design-rule illustration

Every via uses a 0.3 mm drill with a 0.6 mm pad, giving an annular ring of about 0.15 mm on each side — enough copper to reliably anchor the plating after drilling.



Rule 3 — Pad-to-Pad Clearance

Design Rule 3: Pad-to-Pad Clearance



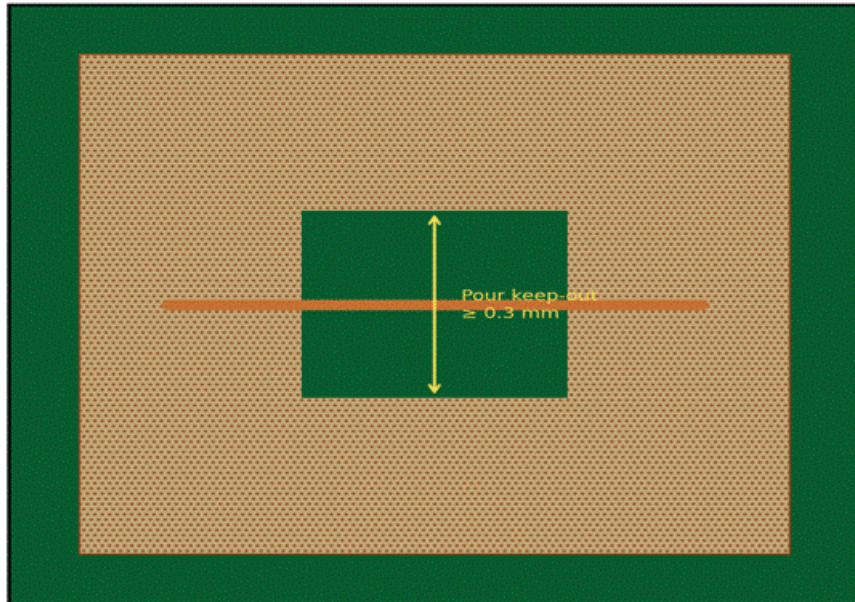
Rule 3 — Pad-to-Pad Clearance — annotated design-rule illustration.

A minimum spacing of 0.2 mm is enforced between neighbouring pads so that solder paste stencils and reflow soldering do not bridge adjacent pins.



Rule 4 — Copper Pour (Ground Fill) Clearance

Design Rule 4: Copper Pour Clearance (Keep-out)



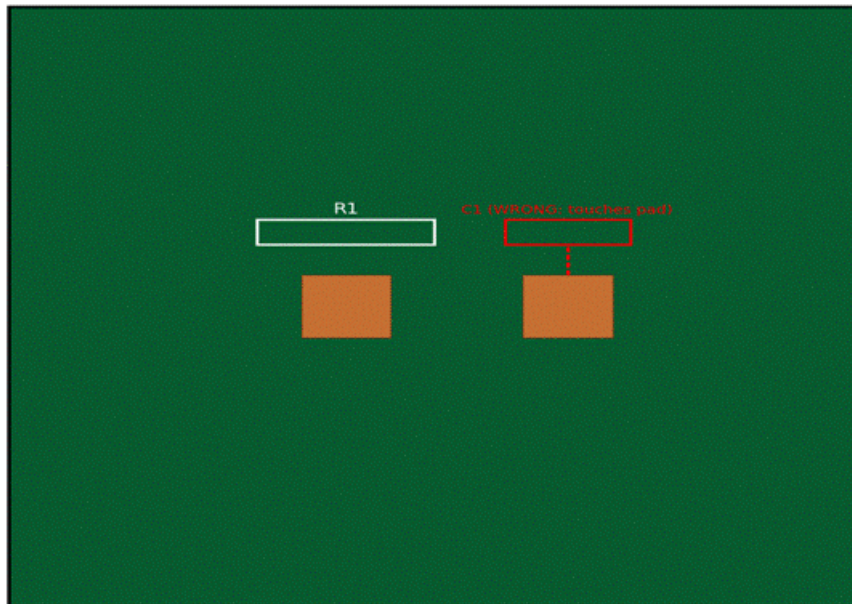
Rule 4 — Copper Pour (Ground Fill) Clearance — annotated design-rule illustration.

The ground copper pour is kept back at least 0.3 mm from any signal trace or pad, preventing accidental shorts while still maximising the return-path copper.



Rule 5 — Silkscreen-to-Pad Clearance

Design Rule 5: Silkscreen-to-Pad Clearance

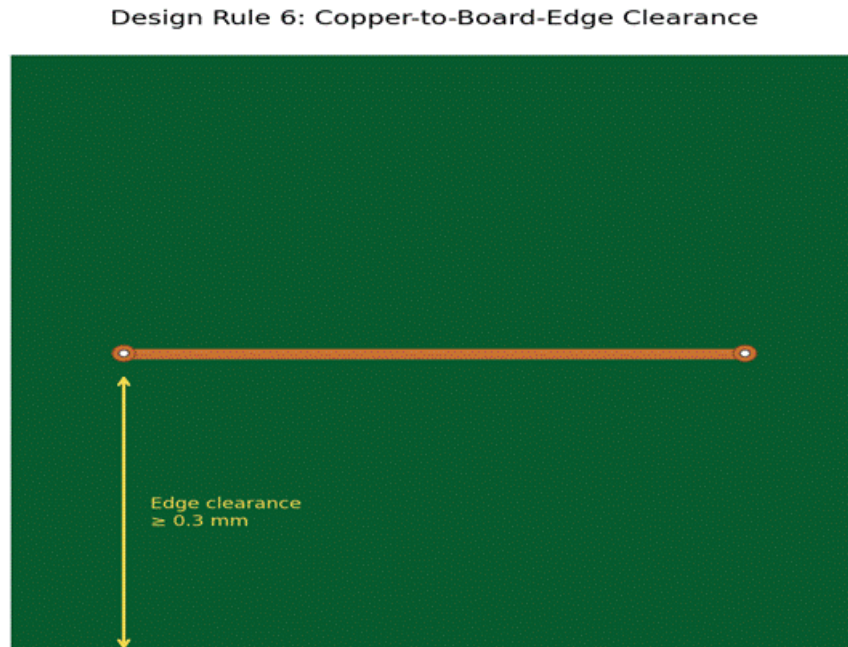


Rule 5 — Silkscreen-to-Pad Clearance — annotated design-rule illustration.

Silkscreen text and outlines are kept clear of exposed copper pads; silkscreen printed over a pad (as shown in red) would contaminate the solder joint and was avoided.



Rule 6 — Copper-to-Board-Edge Clearance



Rule 6 — Copper-to-Board-Edge Clearance — annotated design-rule illustration.

All copper features are kept at least 0.3 mm away from the board outline so that the routing/cutting tolerance during fabrication does not expose or damage copper.

Step 9 — Verifying the +5V Power Net

Once routing was complete, every occurrence of the +5V net was located directly on the schematic and circled in red, shown below: at the USB connector's power-flag symbol, and at the input to the AMS1117 regulator in the 5V-to-3V3 block. Everywhere else on the board — the STM32, the LAN9252, the EEPROM and both RJ45 ports — deliberately runs from the regulated +3.3V rail only.

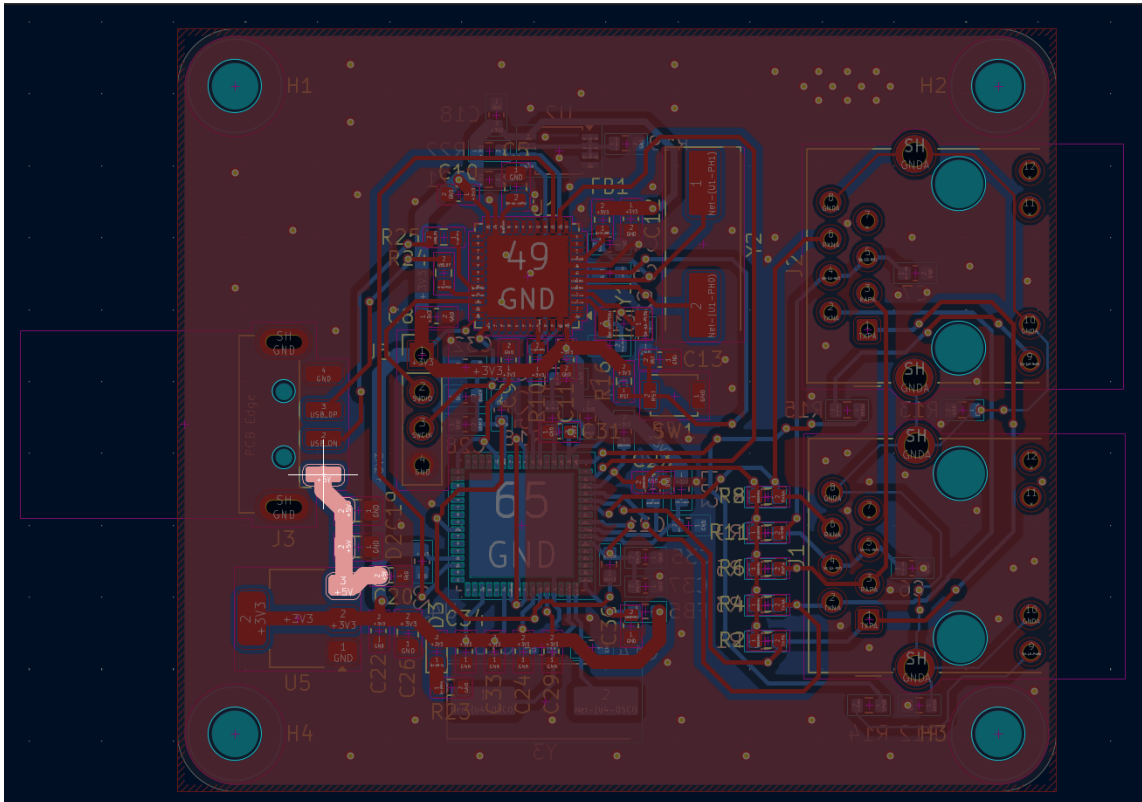


Figure 8: +5V net location is highlighted on the KiCad PCB board — USB VBUS and the AMS1117 regulator input.

This visual net-trace confirmed that +5V reaches exactly two places on the whole schematic — the USB block and the regulator input — and nowhere else, which was important because it meant no 3.3V-rated component could be accidentally connected to the raw 5V rail before fabrication.

Step 10 — Verifying the +3.3V Power Net

The same check was repeated for the +3.3V rail, since this is the supply that actually powers almost every active component on the board. Every "+3V3" label was located and circled in blue: the STM32's decoupling bank and reset pull-up, the LAN9252's TEST/FXSD_EN pins and main VDDIO/VDDCR/VBIAS supply, the AMS1117 regulator output, and the I2C EEPROM's pull-up resistors and decoupling capacitor.

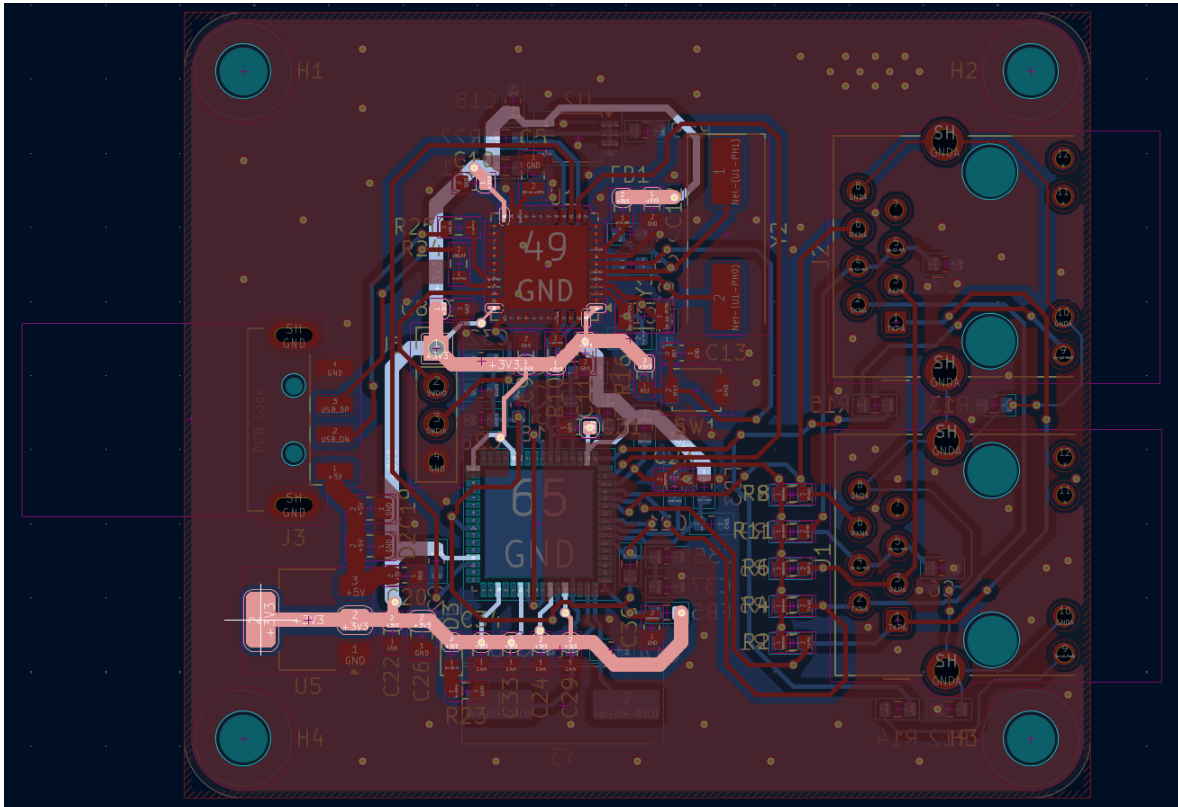


Figure 9: +3.3V net locations is highlighted on the KiCad PCB board — regulator output, STM32 supply pins, LAN9252 control pins, and EEPROM pull-ups.

Unlike the +5V rail, which touches only two points, +3.3V fans out across almost the entire schematic — confirming visually that the STM32, the LAN9252's digital core, and the EEPROM are all correctly referenced to the regulated rail rather than the raw USB supply.

Step 11 — Running the Design Rule Check (DRC)

Before generating the Gerber/fabrication files, KiCad's built-in Design Rule Checker (DRC) was run to automatically verify every rule from Step 8 — clearance, track width, via/hole sizing, annular ring, silkscreen clearance, edge clearance, courtyard overlap, and full net connectivity (no missing ratsnest connections) — across the entire board in one pass.

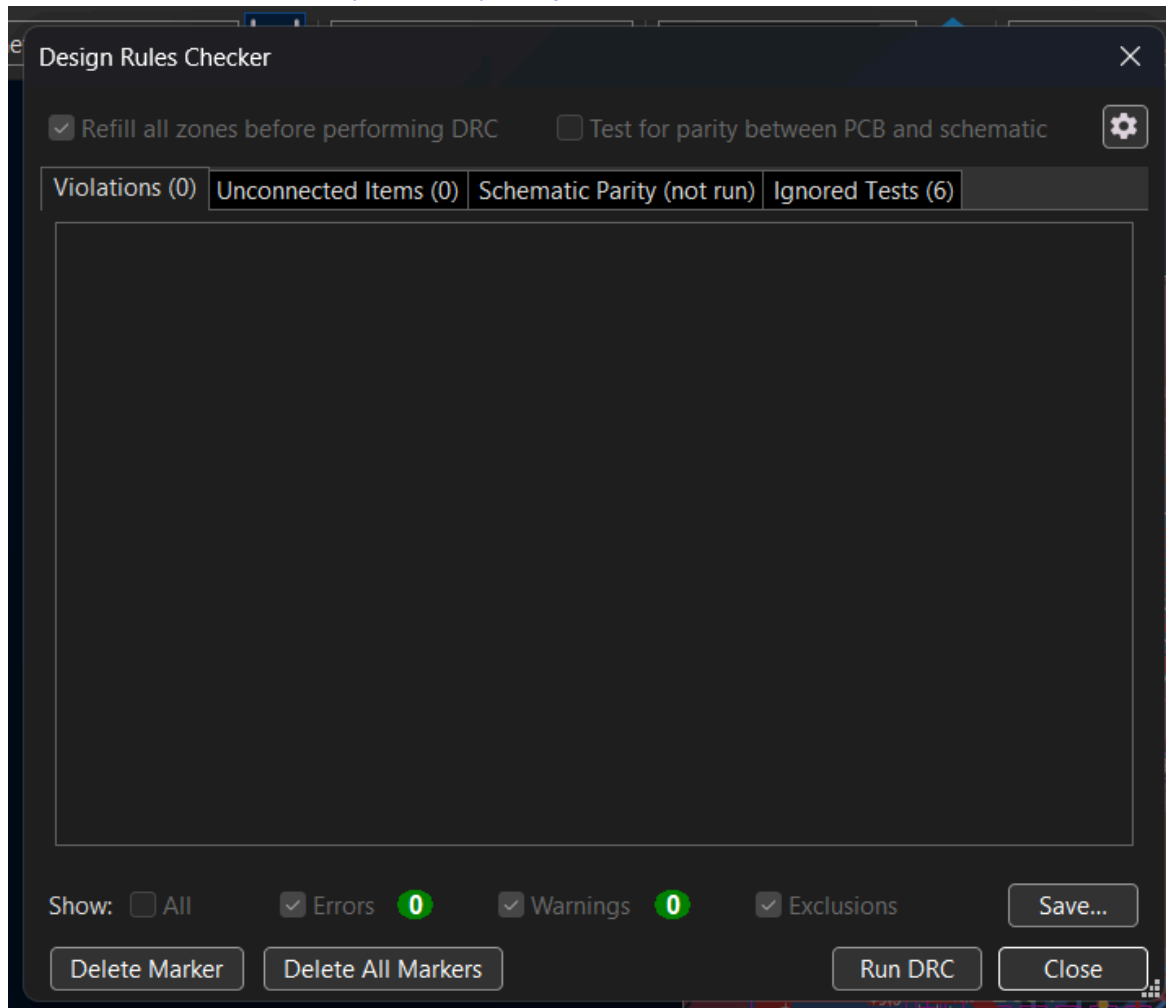


Figure 10: DRC results panel — every rule category passing with 0 violations and 0 unconnected items.

A clean DRC run (0 violations, 0 unconnected items, 0 footprint warnings) was the final gate before this design was considered fabrication-ready; the real board's status bar confirms the same result in numeric form — 334 pads, 151 vias, 559 track segments across 107 nets, and 0 unrouted connections.

Step 12 — 3D Board Preview Before Fabrication

The last check before ordering the boards was KiCad's built-in 3D Viewer, which renders the finished layout with real component and connector models — a last visual sanity check for footprint orientation, connector clearance to the board edge, and overall component fit before committing to fabrication.



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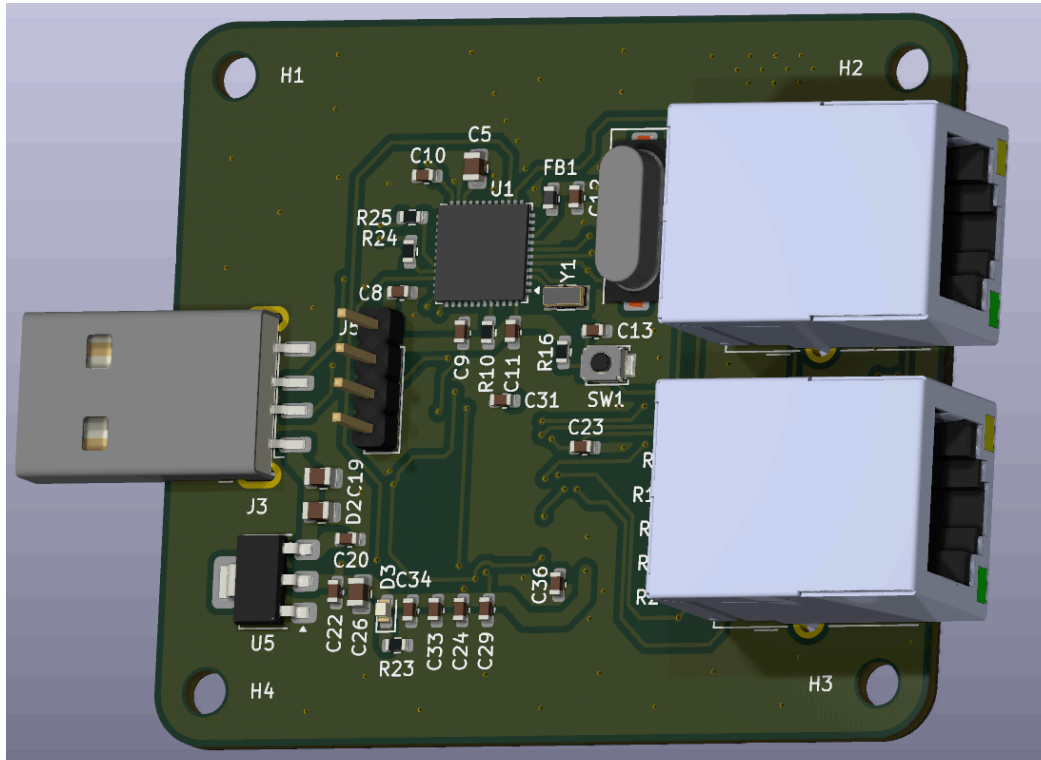


Figure 11: 3D Viewer — top-side render showing the USB connector (J3) and the two stacked RJ45 jacks (J1, J2) in their final positions.

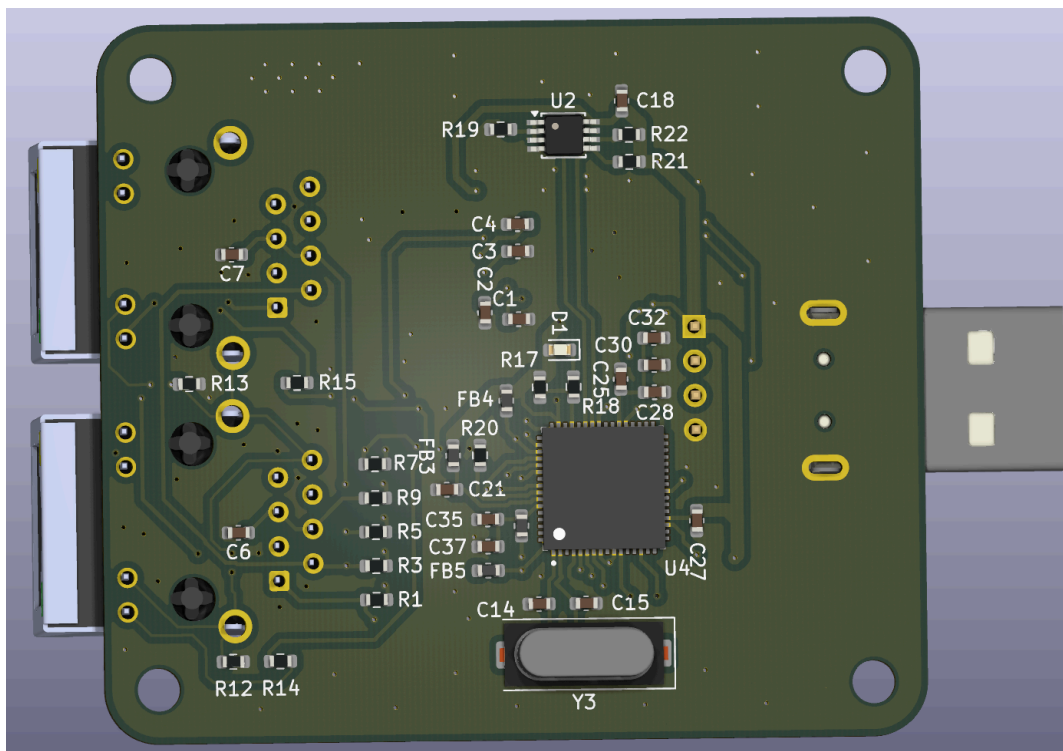
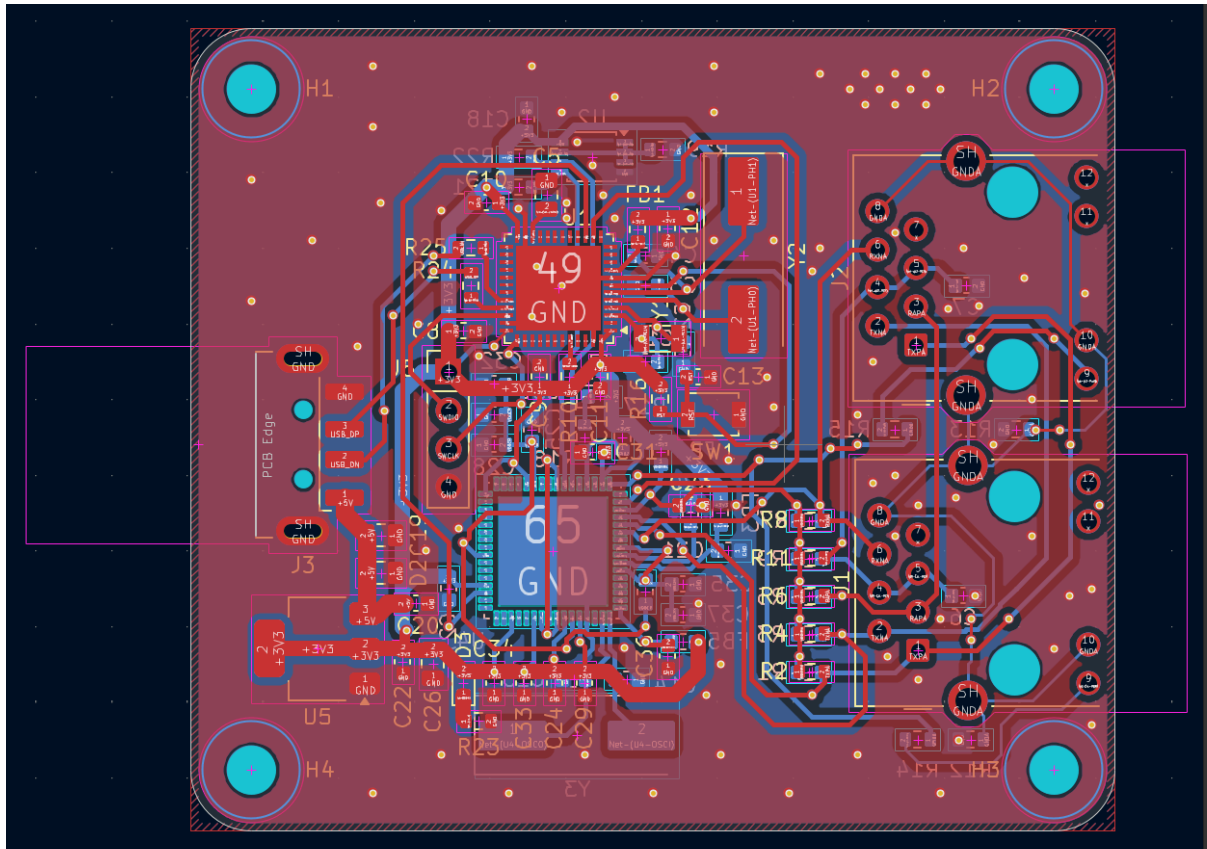


Figure 12: 3D Viewer — the same board viewed from the underside, showing the LAN9252 (U4) and its surrounding passives with the connectors visible through the board silhouette.



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Seeing the connectors fully rendered in 3D — rather than as flat footprint outlines — confirmed there was enough clearance around the USB plug and both RJ45 jacks for real cables to be inserted without interference from the nearby mounting holes or components.



4.7 Project GitHub Link

<https://github.com/Sudeshpandit>



5. Social / Industry Relevance of the Project

Industrial automation increasingly relies on EtherCAT for real-time communication between PLCs, drives and sensors, but engineers, students and small integrators often need a simple, affordable way to bridge a standard PC to an EtherCAT-side circuit for testing, configuration or small-scale deployment. A low-cost USB-to-EtherCAT adapter board fills exactly this gap, making industrial-communication hardware accessible outside large automation labs and giving rural youth entering the electronics and automation industry a directly relevant, portfolio-ready skill.

From an industry-relevance standpoint, the workflow followed here — schematic capture, DRC-driven layout, layer documentation, and bring-up testing in KiCad — mirrors the exact process used by professional hardware design teams, giving the trainee transferable PCB design and industrial-communication skills that map directly onto real embedded-systems and automation job roles.



6. Learning & Reflection

6.1 New Learnings

- Learned end-to-end KiCad schematic capture across mixed digital/analog/high-speed blocks — wiring the STM32's native USB peripheral and oscillator networks, configuring the LAN9252 EtherCAT controller's SPI/I2C interfaces, and laying out its individually-filtered PHY power domains.
- Learned 2-layer routing strategy for a board with sensitive Ethernet PHY signals — keeping the bottom layer mostly as a ground pour beneath the PHY/RJ45 section, and using vias only where necessary to reduce impedance discontinuities.
- Learned how to read and apply a fabrication house's design rules (minimum trace/clearance/drill, annular ring) directly into KiCad's DRC settings, and how to interpret a clean DRC report before ordering boards.
- Learned the practical role of an EtherCAT slave configuration EEPROM (24LC512) and why a dedicated I2C device is needed alongside an EtherCAT controller chip like the LAN9252.

6.2 Personal Experience

Working through this project solo, the transition from schematic to a physically manufacturable board was the most valuable part of the training — concepts like trace width, clearance and annular ring, which initially felt abstract, became concrete once real DRC violations were caused (or avoided) in KiCad. Circling every +5V and +3.3V net directly on the schematic as a review exercise was particularly useful in confirming that the raw 5V rail reached only the intended two points, and that every other IC was correctly referenced to the regulated 3.3V rail — reinforcing the value of methodical power-net review on every future board.



7. Future Scope & Conclusion

7.1 Recap of Objectives and Achievements

The project set out to design, document and fabricate a working 2-layer PCB combining an STM32 microcontroller with a LAN9252 EtherCAT controller, while learning industry-standard PCB design rules in KiCad. All objectives were met: a DRC-clean 2-layer layout was produced, every physical layer was documented, the applied design rules were verified and illustrated (including a dedicated look at vias and the DRC process), the full schematic was explained block by block, both the +5V and +3.3V nets were located and highlighted, and the assembled board was confirmed working through USB enumeration, SPI communication and RJ45 link-up.

7.2 Future Scope

Migrate to a 4-layer stack-up with a dedicated ground and power plane to further improve signal integrity for the LAN9252's high-speed Ethernet PHY lines.

- Add ESD/EMI protection components on the USB and EtherCAT-facing lines for industrial-environment robustness.
- Design a matching enclosure and panelise the board for small-batch production runs.

In conclusion, this project gave the trainee practical, end-to-end exposure to professional PCB design practice in KiCad — from schematic to a tested, working board — and produced a genuinely useful USB-to-EtherCAT adapter platform that can be extended in future training cycles.